

minimizing risk. It's an approach that would make any math teacher smile. But there is an important assumption embedded in the approach: the inputs, which are based on history, amount to a forecast of returns, correlations, and volatility for the future.

Although Markowitz did win a Nobel Prize, at least partly based on his elegant mathematical solution to identifying mean-variance efficient portfolios, a funny thing happened when his ideas were applied in the real world: mean-variance performed poorly.

The reason is that the *historical* returns, and correlations, which informed the input for the optimization algorithm, were *different* from what investors actually experienced in the real world in the future. That is, the estimates were “unstable,” in the sense that they turned out to be guesses that simply weren't any good.

Academics began to see the basic flaw in the approach: Estimates for any covariance matrix are going to be unstable, and thus the “optimized” weights spit out from a model influenced by an unstable covariance matrix will also end up being unstable and unreliable. And so this highly engineered, mathematically beautiful, and Nobel Prize–winning approach translated into a no-value-added-situation for investors.

This is something to keep in mind when considering any optimization method for asset allocation. It also emphasizes a critical point related to asset allocation that every investor should be aware of: *Complexity does not equal value*.

Recall Charlie Munger's admonition to “invert, always invert.” So if complexity does not equal value, is it perhaps simplicity that equals value? Later in [Chapter 6](#), we will discuss how simple equal-weight allocations do indeed seem to reliably beat complex allocation approaches.

Markowitz himself may have realized the value of simplicity intuitively. Consider the irony: The founder of modern portfolio theory uses an equal-weight allocation. And one of the central assumptions underlying mean-variance optimization is that investors care about risk and return trade-offs. Yet, as Markowitz highlights, his decision-making framework has little to do with risk and return trade-offs. In 2014, now that we have a long enough data trail, we can show that Markowitz's model doesn't outperform a simple